CS-30003 · Coding Assignment 1 · Project Proposal

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Section	CS-33
Repository	https://github.com/sneharam543/shardswarm
Project	P8 · P2P File Distribution (BitTorrent-lite)
Path	A (Catalogue)

1. WHAT WE ARE BUILDING

A BitTorrent-lite peer-to-peer file distribution system built from raw TCP sockets, consisting of a custom tracker, a self-specified peer wire protocol, and a multi-peer swarm client. It sits at the application/transport boundary of the network stack: unlike a real torrent client, every component (metainfo format, tracker, wire protocol, piece selection, and choking) is implemented by us rather than consumed from a library.

2. CORE DELIVERABLES

- Metainfo format with per-piece SHA-256 hashes.
- Tracker (HTTP or UDP) maintaining swarm membership and serving peer lists on request.
- Self-specified, documented peer wire protocol over raw TCP: handshake, bitfield, have, request, piece, cancel.
- Rarest-first piece selection with request pipelining and per-piece hash verification on receipt.
- Live multi-peer swarm: at least 6 concurrent peers with 1–2 seeders, run as independent processes.

3. THE CLAIM WE WILL TEST

Claim: A rarest-first piece selection policy combined with tit-for-tat choking completes distribution faster and penalizes non-contributing peers more effectively than random or sequential selection with no choking. **Experiment:** fix swarm size (6 peers, 1–2 seeders) and file size as controls; vary piece-selection strategy (rarest-first / random / sequential) and choking policy (on / off) as independent variables; measure total distribution time, per-peer download rate, and completion time of an injected free-rider peer as metrics. Bandwidth is deliberately rate-limited per peer (token-bucket) since all peers run on shared hardware. Results presented as distribution-time-vs-strategy bar charts and download-rate-vs-time line plots (matplotlib/pandas).

4. STACK AND JUSTIFICATION

Go for the tracker, peer client, and wire protocol, since goroutines and channels map naturally onto one state machine per peer connection, which is essential for genuine (not simulated) concurrency across 6 peers. **SHA-256** (standard crypto library) for metainfo and piece hashing. **Docker Compose** to run tracker + peers as independent containers/processes for a realistic swarm demo. **matplotlib / pandas / numpy** for experiment analysis and plotting (explicitly permitted as supporting libraries).

5. WORK SPLIT

Sneha Ram (Metainfo & Tracker): metainfo format, per-piece SHA-256 hashing, tracker server, swarm membership, peer-list serving.
Aditya Giri (Peer Wire Protocol): handshake and message framing (bitfield/have/request/piece/cancel) over raw TCP; owns the written protocol specification.
Rishav Sarkar (Download Engine & Concurrency): rarest-first piece selection, request pipelining, on-receipt hash verification, per-peer state-machine correctness across the 6-peer swarm.
Swayam Kanjilal (Choking & Experiments): choking/tit-for-tat logic, free-rider peer and experiment, rate-limiting design, and all plotted experiment analysis.

6. RISKS

- **Fake concurrency:** peers implemented as sequential loops instead of independent state machines. Mitigation: Rishav builds and unit-tests the per-peer state machine first, before swarm-scale integration.
- **Invalid single-machine measurements:** unthrottled peers on one host produce meaningless bandwidth numbers. Mitigation: Swayam implements token-bucket rate limiting before any experiment data is collected.

7. WEEKLY PLAN

Week	Target	Owner
1	Repo setup, metainfo format design, protocol spec draft	Sneha / Aditya
2	Tracker (announce/peer-list) + peer handshake implementation	Sneha / Aditya
3	Full wire protocol messaging (bitfield, have, request, piece, cancel)	Aditya
4	Piece selection engine + per-peer state machine (2 peers)	Rishav
5	Scale to 6-peer swarm, hash verification, request pipelining	Rishav
6	Choking logic + free-rider peer implementation	Swayam
7	Rate-limiting, full experiment suite (selection strategy, free-rider)	Swayam / All
8	Integration, plots, report, viva prep	All